Shared 3-s LSTM Forecasting for Coupled MPC–DWA UAV Planning in Dynamic Obstacle Fields

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***Abstract*—First, three-dimensional UAV planning around moving obstacles is difficult because turns, acceleration and pauses make future encounter positions uncertain. Route planning and local action screening can also disagree when they use different obstacle futures. Second, we couple model predictive control (MPC) and the dynamic window approach (DWA) through one causal long short-term memory (LSTM) forecast. The LSTM predicts 3 s of obstacle motion; MPC uses it across the route horizon, while DWA tests reachable actions against the same positions and bounds MPC's first command. Third, in the five integrated arms (1,500 episodes) of a 2,100-episode matched simulation bank, shared LSTM reached without collision in 221/300 runs (73.7%) with no swept contacts. It exceeded latest-position hold by 16.3 percentage points (simultaneous 95% scene-cluster confidence interval, 5.1–27.6) and coupled constant velocity by 5.7 points (-3.4–14.7). Differences from one-layer LSTM injection were at most 1.0 point, and both intervals included zero. Finally, shared LSTM improved collision-free arrival over latest-position hold in these simulations; differences from coupled constant velocity and either one-layer LSTM variant remained uncertain. Flight validation remains future work.**

***Index Terms*—UAV, 3D path planning, dynamic obstacles, LSTM, model predictive control, dynamic window approach**

I. Introduction

Unmanned aerial vehicles (UAVs) must choose safe motion before a moving obstacle reaches the same space. Recent positions are poor guides when an obstacle turns, accelerates or pauses. Model predictive control (MPC) can plan a short future route, while the dynamic window approach (DWA) can select an action that the vehicle can reach immediately [1], [2]. Forecast-based UAV planners have used predicted obstacle motion in route optimization [3], [4], but a coupled planner also needs its route and local-action layers to reason about the same future.

We address this interface with one 3-s LSTM forecast shared by both layers. MPC uses the forecast to shape its route, DWA uses the same time-aligned positions to screen reachable actions, and the range of screened DWA controls constrains MPC's next first command. A trust score then shifts DWA sampling between the MPC guide and exploration. We formulate this forecast-sharing interface within a coupled MPC–DWA planner and compare it with coupled constant-velocity (CV) prediction, single-layer LSTM injection and latest-position hold.

II. Problem Formulation

The planning scene is a finite three-dimensional workspace with a start and a goal. The UAV and each moving obstacle are represented by spheres, and each obstacle follows a trajectory generated before the run, independently of the UAV. The planner observes the UAV's current state and the obstacles' current states and histories, but cannot access their future true positions. Figure 1 defines this geometry; recorded planner paths are shown separately.

At each replanning step, the planner chooses a command containing desired speed, pitch rate and yaw rate. It predicts motion over a rolling horizon but applies only the next update's screened action before observing the scene again and replanning. The task is to reach the goal while remaining inside the flight corridor and avoiding contact with every obstacle along the trajectory. An episode ends at goal arrival, when the planner has no screened action to issue, or at the simulator's stopping limit.

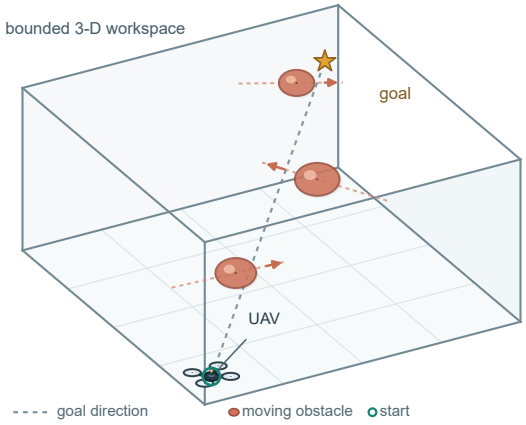


Figure 1. Problem geometry: the quadrotor is centered at the start and moves toward the goal among spherical obstacles following prescribed paths (arrows). The dashed line marks the direct start-to-goal direction; planner trajectories appear in Figure 4.

III. Planning Method

Figure 2 follows one planning update. The LSTM first converts recent obstacle states into a time-aligned 3-s sequence.

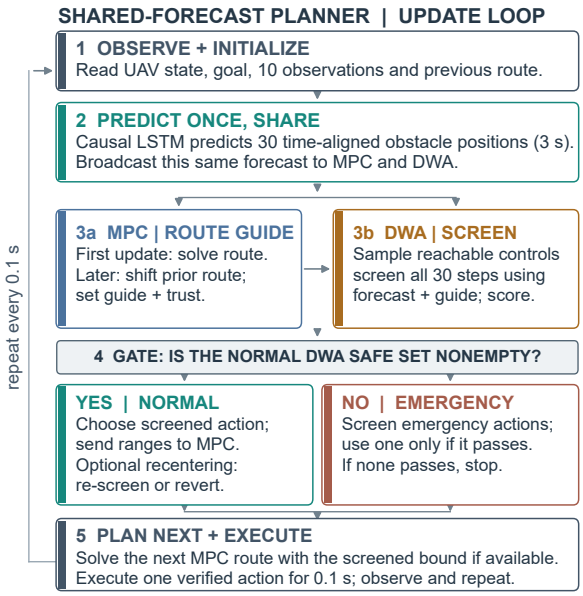


Figure 2. Five modules in one 0.1-s planning update. One 3-s LSTM forecast feeds the MPC route plan and DWA rollout screen. The MPC guide shapes DWA sampling, and the screened DWA set bounds MPC's next first command. Normal, recentered and emergency actions are screened before execution.

At startup, MPC uses this sequence to create an initial route; later updates shift the preceding MPC solution forward as the new guide. The guide sets a trust score, which divides DWA samples between controls near the guide and controls that explore other directions. DWA rolls each sample forward for 3 s, rejects candidates that violate clearance, corridor or braking checks, and scores the survivors.

The safe candidates also send information back to MPC. Their component-wise control ranges form a box that constrains only the first command of the new MPC plan; the box itself does not certify every command inside it. The new plan guides the next update, while the separately screened best DWA action is executed now. If optional lateral recentering changes that action, the modified rollout is screened again. If the normal set is empty, emergency actions pass a second safety check; the simulator stops when none passes. Thus an unchecked MPC or emergency command is never applied.

A. *Core equations*

The equations below define the vehicle update, obstacle forecast and planner decisions. The vehicle state contains position, speed, pitch and yaw. The signed distance between its collision sphere and obstacle j is

$$\begin{aligned}
\mathbf{h}_t &= [\cos\theta_t\cos\psi_t, \cos\theta_t\sin\psi_t, \sin\theta_t]^\mathsf{T},\\
\mathbf{p}_{t+1} &= \mathbf{p}_t + \Delta t\, v_t \mathbf{h}_t,\\
v_{t+1} &= v_t + \frac{\Delta t}{\tau_v}(v_{c,t} - v_t),\\
\theta_{t+1} &= \theta_t + \Delta t\,\omega_{\theta,t}, \quad \psi_{t+1} = \psi_t + \Delta t\,\omega_{\psi,t},\\
\hat{d}_{j,k} &= \|\mathbf{p}_{t+k|t} - \hat{\mathbf{o}}_{j,t+k|t}\|_2 - r_u - r_j.
\end{aligned} \tag{1}$$

Here t is the current step, k a future step, and $j = 1, \ldots, M$ indexes obstacles. The unit direction $\mathbf{h}_t$ follows pitch θ_t and yaw ψ_t. The update period is $\Delta t = 0.1$ s; τ_v is the speed-response time; and $v_c, \omega_\theta, \omega_\psi$ are commanded speed and angular rates. In the clearance term, $\mathbf{p}_{t+k|t}$ and $\hat{\mathbf{o}}_{j,t+k|t}$ are predicted positions. The radii are r_u and r_j. Positive $\hat{d}_{j,k}$ means separation, zero contact and negative overlap.

At each update, the causal LSTM [5] maps ten obstacle observations to thirty 3-D offsets from the latest observed centre:

$$\begin{aligned}
\mathbf{q}_{j,s} &= [\mathbf{o}_{j,s}^\mathsf{T}, \mathbf{v}_{j,s}^\mathsf{T}, \cos\chi_{j,s}, \sin\chi_{j,s}]^\mathsf{T},\\
\hat{\mathbf{o}}_{j,t+k|t}^{\mathrm{LSTM}} &= \mathbf{o}_{j,t}^{\mathrm{obs}} + [f_\Theta(\mathbf{q}_{j,t-9:t})]_k.
\end{aligned} \tag{2}$$

Here s indexes history and $k = 1, \ldots, 30$ indexes forecast steps. The observed position $\mathbf{o}$ and velocity $\mathbf{v}$ are followed by cosine and sine of heading χ, giving eight input features. The trained network f_Θ outputs one 3-D offset per future step, and $[\cdot]_k$ selects offset k. Ten observations span 0.9 s and the forecast spans 3 s. Each offset is anchored to the latest observed centre, not cumulatively summed. At startup, the earliest state fills missing history. Constant-velocity (CV) baselines extrapolate from the latest two positions, using zero initial velocity when only one is available; position hold repeats the latest centre. Training details are in Supplementary Methods.

MPC uses the forecast over its 30-step horizon to minimize goal error, control effort and changes, route deviation, and obstacle-clearance penalties:

$$\begin{aligned}
\min_{\mathbf{u}_{1:30}} J_{\mathrm{MPC}} = &\sum_{k=0}^{29} w_p \|\mathbf{p}_{t+k|t} - \mathbf{g}\|_2^2\\
&+ w_f \|\mathbf{p}_{t+30|t} - \mathbf{g}\|_2^2 + w_r J_r\\
&+ \sum_{k=1}^{30} w_u \|\mathbf{u}_k\|_2^2 + w_{\Delta u} \sum_{k=2}^{30} \|\mathbf{u}_k - \mathbf{u}_{k-1}\|_2^2\\
&+ \sum_{k=1}^{30} \sum_j \left(w_o [-\hat{d}_{j,k}]_+^2 + w_s [d_{\mathrm{soft}} - \hat{d}_{j,k}]_+^2 \right),\\
&\text{s.t. } \mathbf{x}_{t+k|t} = f(\mathbf{x}_{t+k-1|t}, \mathbf{u}_k), \quad \mathbf{u}_1 \in \mathcal{U}_t \text{ if } \mathcal{A}_t \neq \varnothing.
\end{aligned} \tag{3}$$

Here $\mathbf{u}_{1:30}$ is the planned command sequence, $\mathbf{x}$ is the position–pitch–yaw–speed state, $\mathbf{g}$ is the goal, and f is the state transition. Each command is $\mathbf{u}_k = [v_{c,k}, \omega_{\theta,k}, \omega_{\psi,k}]^\mathsf{T}$, containing desired speed, pitch rate and yaw rate. The weights $w_p, w_f, w_r, w_u, w_{\Delta u}, w_o, w_s$ penalize, respectively, stage goal error, terminal goal error, route deviation, command magnitude, command changes, predicted overlap and clearance below $d_{\mathrm{soft}} = 0.35$ m. Here J_r is the sum of squared cross-track deviations from the direct start-to-goal line. The positive-part operator is $[x]_+ = \max(0, x)$. These are optimization costs; overlap and clearance are soft penalties. The vehicle dynamics, physical command limits and per-step command-change limits remain hard constraints throughout the horizon. When DWA has safe candidates, only MPC's first command is additionally constrained to their expanded control box. If the normal safe set is empty, MPC uses its ordinary physical bounds to update the next guide, while execution remains with the screened DWA or emergency action. Full definitions and limits are in Supplementary Methods S1.1.

Trust allocates DWA samples between guided and exploratory actions using safety, goal progress, agreement with the previous MPC guide, and control smoothness:

$$\begin{aligned} \tau &= \sigma_s \tau_{\mathrm{sa}} + \sigma_g \tau_g + \sigma_t \tau_{\mathrm{ref}} + \sigma_m \tau_{\mathrm{sm}}, \\ \tau_{\mathrm{ref}} &= \exp(-\alpha_T \Delta_p), \\ B &= \mathrm{round}[300(1 + \alpha e^{-\beta\tau})], \\ B_g &= \mathrm{round}(B\tau), \quad B_e = B - B_g. \end{aligned} \tag{4}$$

The four trust components describe the current MPC guide. Safety is its minimum predicted clearance. Goal progress is the decrease in goal distance from the guide's first to last point. Reference match falls with the distance Δ_p between the UAV and the guide's first point, and smoothness measures changes between planned controls. All four terms are normalized to $[0, 1]$. The coefficients $\sigma_s, \sigma_g, \sigma_t, \sigma_m$ weight safety, progress, reference match and smoothness, respectively, and sum to one. The decay α_T sets how quickly reference match falls with distance; α and β set the size and trust sensitivity of the sampling budget. Here B_g and B_e are the guided and exploratory candidate counts. Higher trust sends more samples toward the MPC guide while reducing the total count. Six coefficient vectors were tested across five integrated arms, 48 development scenes and three seeds ($6 \times 5 \times 48 \times 3 = 4{,}320$ episodes). The selected vector $[0.15, 0.15, 0.15, 0.55]$ produced the highest equal-arm mean safe-arrival rate, 0.8042, and was fixed before the 100-scene evaluation. Full component definitions are in Supplementary Methods S1.2.

DWA rejects unsafe rollouts, scores the survivors, and uses their control range to bound MPC's first input:

$$\begin{aligned} \mathcal{A}_t &= \{b \in \mathcal{B}_t : d_{\min}^{(b)} > 0.25 \text{ m},\ \mathcal{C}_b = 1, \\ &\qquad (v_{\mathrm{end}}^{(b)})^2 \le 2 a_{\max} d_{\min}^{(b)}\}, \quad b^* = \arg\max_{b \in \mathcal{A}_t} S_b, \quad \mathcal{A}_t \neq \varnothing, \\ u_{i,t}^{-} &= \max\left(u_i^{\min}, \min_{b \in \mathcal{A}_t} u_i^{(b)} - \delta\right), \\ u_{i,t}^{+} &= \min\left(u_i^{\max}, \max_{b \in \mathcal{A}_t} u_i^{(b)} + \delta\right), \\ \mathcal{U}_t &= \prod_{i=1}^{3} [u_{i,t}^{-}, u_{i,t}^{+}], \quad \delta = 0.1. \end{aligned} \tag{5}$$

Here b indexes a candidate, $\mathcal{B}_t$ is the sampled set, and $\mathcal{A}_t$ contains candidates passing all three gates. The minimum surface clearance $d_{\min}^{(b)}$ is checked at all 30 time-aligned rollout states against the shared forecast and must exceed 0.25 m. The indicator $\mathcal{C}_b = 1$ means the whole rollout stays inside the altitude limits and route corridor. Its terminal speed is $v_{\mathrm{end}}^{(b)}$, and $a_{\max} = 1.0$ m s^{-2} is the available braking deceleration. The score S_b favors agreement with the MPC guide, obstacle clearance and a suitable vertical profile. For a nonempty safe set, i denotes desired speed, pitch rate or yaw rate. The interval in each channel expands the observed safe range by $\delta = 0.1$ in that channel's native units, then clips it to the physical limits $u_i^{\min}$ and $u_i^{\max}$. Their product $\mathcal{U}_t$ constrains MPC's first input. A continuous piecewise-linear audit checks between saved simulator states when outcomes are classified. Full scores and fallback rules are in Supplementary Methods S1.2–S1.4.

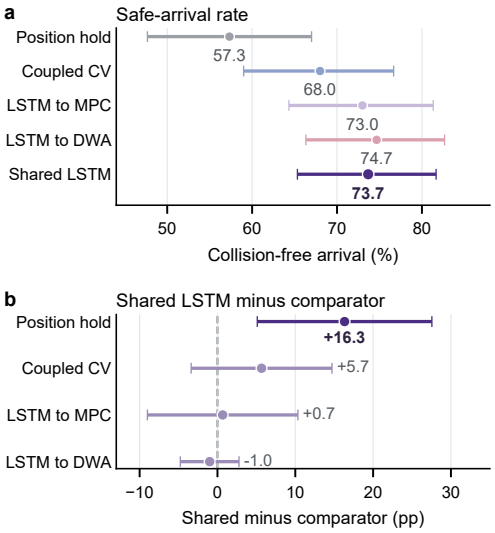


Figure 3. Integrated-controller results with a 3-s MPC and DWA horizon. (a) Collision-free arrival rate with scene-bootstrap percentile 95% CIs ($n = 300$ per method; 100 scenes and three matched seeds). (b) Shared-LSTM minus comparator differences with simultaneous max-$|t|$ 95% scene-cluster CIs; the four integrated contrasts are displayed, while multiplicity control retains all six original contrasts. Positive values favor sharing. Position hold recorded 50 swept-contact episodes, while the other four methods recorded none. Cross-family standalone results are reported in Supplementary Tables S11–S12.

B. How the shared forecast is used

The LSTM does not command the UAV. It supplies future obstacle centres on the same time grid to two different decisions. MPC uses all 30 positions in its clearance costs, so the route can bend before a future encounter. DWA compares each reachable rollout with those same positions, so the next action is checked against the future assumed by MPC. The preceding MPC guide also changes DWA's guided/exploratory sample mix, while the accepted DWA range limits MPC's next first command. This shared input plus two-way feedback is the central design change. Supplementary Methods S1.4 gives the complete execution order.

IV. Simulation and Results

We next test whether the shared forecast improves a coupled planner when the prediction horizon, controller horizon, scenes and random seeds are matched.

The complete exploratory bank contains 2,100 episodes over seven arms. Here we focus on the five integrated MPC–DWA arms, giving 1,500 episodes, while retaining the original six-contrast simultaneous correction; the two standalone arms remain in Supplementary Tables S11–S12. Each arm used 0.1-s updates, 3-s MPC and DWA horizons, 100 scenes, and three matched seeds per scene. In the one-layer LSTM variants, the other layer used CV extrapolation; coupled CV and position hold supplied their respective predictions to both layers. A safe arrival required goal tolerance with no swept-path contact. Seeds were averaged within scene, and 20,000 scene-cluster bootstrap resamples formed simultaneous intervals for shared-minus-comparator differences.

Shared LSTM reached safely in 221/300 runs (73.7%). Latest-position hold reached 172/300 (57.3%), giving a shared-forecast gain of 16.3 percentage points (95% CI, 5.1–27.6). Coupled CV reached 204/300 (68.0%); the corresponding point estimate was 5.7 points higher (−3.4–14.7). LSTM supplied only to MPC reached 219/300 (73.0%), and LSTM supplied only to DWA reached 224/300 (74.7%). Their differences from sharing were +0.7 points (−9.0–10.3) and −1.0 point (−4.8–2.8), respectively. Shared LSTM therefore improved collision-free arrival over position hold; the intervals did not establish an advantage over coupled CV or either one-layer LSTM variant (Fig. 3).

Table 1
INTEGRATED 3-S OUTCOMES. SAFE ARRIVAL REQUIRES NO SWEPT CONTACT.

Method	Safe arrival	Swept contact
Position hold	172/300 (57.3%)	50
Coupled CV	204/300 (68.0%)	0
LSTM to MPC	219/300 (73.0%)	0
LSTM to DWA	224/300 (74.7%)	0
Shared LSTM	221/300 (73.7%)	0

Figure 4 shows one matched case selected post hoc for outcome and route visibility. Only shared LSTM reached safely among the five integrated methods.

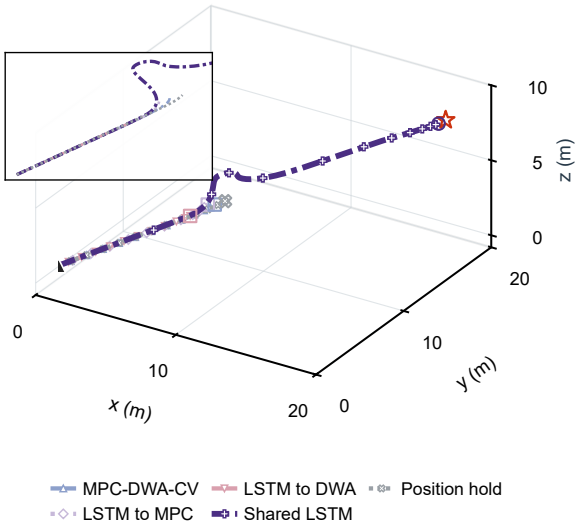


Figure 4. Five integrated paths in one matched case (scene 72, seed 2); the inset enlarges early XY motion. The black triangle and red star mark the start and goal. Only shared LSTM reached safely; coupled CV and the two one-layer LSTM variants stopped without contact, while position hold had a swept contact. Lines stop at logged endpoints. The circle, squares and cross mark safe arrival, other non-arrival and a contact episode; the cross reports the outcome, not the contact location.

V. DISCUSSION

The two control layers solve different parts of the same encounter. MPC looks far enough ahead to bend the route before a predicted conflict, while DWA checks whether the next reachable action passes local safety gates. Giving them the same 3-s obstacle sequence removes an avoidable source of disagreement. The expanded component-wise box constrains MPC's next first command using the ranges of screened DWA controls; it does not certify every command inside, and the separately screened DWA action remains the one executed.

This division also keeps each module's role explicit. The LSTM estimates obstacle motion once per update but does not choose a vehicle command. MPC turns those estimates into an anticipatory route, and DWA tests what the UAV can reach immediately. Returning only the range of screened controls lets local feasibility shape the next route optimization while the selected DWA action remains the command that is executed. The interface can therefore be inspected at the forecast, route, candidate and executed-action stages.

The outcome pattern is consistent with this design. Position hold supplies no obstacle motion, so both layers can act on a rapidly outdated picture; sharing the forecast removes that limitation and improved collision-free arrival. Coupled CV already represents motion, and each one-layer LSTM variant still gives one planning layer a learned forecast, so the remaining point differences were smaller. In the tested scenes, their intervals did not establish an additional arrival benefit from supplying LSTM forecasts to both layers. The method was evaluated with prescribed obstacle tracks, point forecasts and simulator-recorded motion features. Prediction uncertainty, unseen motion families and real flight remain the next validation steps.

VI. CONCLUSION

The planner shares one 3-s LSTM forecast with MPC and DWA, then feeds DWA's screened control range back to MPC. In simulation it achieved 73.7% collision-free arrival, 16.3 percentage points above position hold. Differences from coupled CV and one-layer LSTM injection remained uncertain; flight performance remains to be measured.

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